# Clique packing densities in graphs with forbidden cliques

Jialin He*

**Abstract**

For each fixed $r \geq 3$, we determine the asymptotic minimum number of edge-disjoint copies of $K_r$ in a $K_{r+1}$-free graph at every feasible edge density. The density bound is linear near the Turán threshold and has an explicit nonlinear branch at higher densities. It holds for fractional packings without an error term and for integer packings up to $o(n^2)$, with sharpness for every prescribed sequence of feasible edge counts. For $K_4$-free graphs, the same curve gives a finite integer triangle-packing bound attained by complete tripartite graphs; it sharpens the bound of Győri and Keszegh above density 5/16. We also prove that every $n$-vertex $K_5$-free graph satisfies $\nu_3^*(G) \geq e(G) - \lfloor n^2/4 \rfloor$, and determine its asymptotic triangle-packing density as $\max\{0, p - 1/4\}$ when $e(G)/n^2 \to p$. The proofs use fractional symmetrization and an optimization of the greedy-partition argument for triangles.

**Key words.** Clique packings, forbidden cliques, fractional packings, extremal graph theory.
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## 1 Introduction

A basic packing problem asks how many edge-disjoint cliques a graph must contain in terms of its order and size. For a graph $G$, let $\nu_r(G)$ denote the maximum number of pairwise edge-disjoint copies of $K_r$ in $G$. We determine the sharp asymptotic dependence on edge density when $K_{r+1}$ is forbidden, and give a finite integer bound for triangles. All graphs are finite and simple. We write $T_j(n)$ for the balanced complete $j$-partite graph on $n$ vertices and $t_j(n) = e(T_j(n))$. A forbidden subgraph need not be induced. Turán's theorem places the threshold for forcing a $K_r$ at $t_{r-1}(n)$.

Triangle packings have long been studied through edge decompositions. Erdős, Goodman, and Pósa [4] proved that every $n$-vertex graph can be decomposed into at most $t_2(n)$ triangles and single edges. Counting the members of such a decomposition gives $\nu_3(G) \geq (e(G) - t_2(n))/2$. Král', Lidický, Martins, and Pehova [12] later obtained a decomposition into edges and triangles whose total order is at most $n^2/2 + o(n^2)$. Blumenthal, Lidický, Pehova, Pfender, Pikhurko, and Volec [3] determined the sharp bound on this minimum total order and characterized the extremal graphs for all sufficiently large $n$. In particular, their upper bound $n^2/2 + 1$ yields $\nu_3(G) \geq (2/3)(e(G) - t_2(n)) - O(1)$ for arbitrary graphs.

The $K_4$-free problem admits a stronger conclusion. Motivated by Erdős's question on clique decompositions with cost $|V(C)| - 1$ for each member $C$, Győri [6, p. 360, Conjecture 4] recorded the conjecture that every $K_4$-free graph with $t_2(n) + k$ edges contains at least $k$ edge-disjoint triangles.

*School of Mathematical Sciences, Key Laboratory of MEA (Ministry of Education), Shanghai Key Laboratory of PMMP, and Nantong Institute for Applied Mathematics and Artificial Intelligence, East China Normal University, Shanghai 200241, China. Emails: jlhe@math.ecnu.edu.cn. Supported in part by the National Natural Science Foundation of China (Grant No. 12601663) and the Science and Technology Commission of Shanghai Municipality (Grant No. 22DZ2229014).

In this graph class every clique has order at most three, and a decomposition containing $m$ triangles has cost $e(G) - m$, so the assertion is equivalent to a cost bound of $t_2(n)$. Huang and Shi [11] established quantitative partial bounds, including the conjectured bound in a range of excesses proportional to $n^2$. They also introduced a coloring argument that converts a triangle count into a packing by means of a partition into cliques. Győri and Keszegh [7] resolved the conjecture by combining this argument with a triangle-counting inequality for greedy clique partitions. They also asked for a stronger bound at higher densities.

For general graphs, Balogh, He, Krueger, Nguyen, and Wigal [1, Theorems 1.4, 1.7, and 1.8(ii)] proved that every $n$-vertex graph admits, under this cost function, an edge decomposition of total cost at most $n^2/4 + o(n^2)$, and a fractional edge decomposition of cost at most $t_2(n)$. The latter gives $\nu_3^*(G) \geq e(G) - t_2(n)$ when $G$ is $K_4$-free, where $\nu_3^*$ denotes the fractional packing number defined below. In a recent preprint, Zhang [14, Theorem 1.4] proved the exact integer cost bound $t_2(n)$ for all sufficiently large graphs.

For arbitrary graphs, Balogh and Wigal [2] proved that $e(G) = t_{r-1}(n) + k$ guarantees $\nu_r(G) \geq (2/r - o(1))k$ for every fixed $r \geq 3$. A theorem of Győri [5], in the form stated in [2, Theorem 1.6], gives the stronger estimate $\nu_r(G) \geq k - O_r(k^2/n^2)$ when $k = o(n^2)$. Our question concerns the full density range under a forbidden-clique condition. The extremal packing density agrees with the excess near the Turán threshold, but becomes nonlinear above an explicit transition point.

A fractional $K_r$-packing assigns a nonnegative weight to each copy of $K_r$, with total weight at most 1 on every edge. We denote its maximum total weight by $\nu_r^*(G)$. Haxell and Rödl [8] proved that, for each fixed $r$, $\nu_r^*(G) - \nu_r(G) = o(n^2)$ uniformly over all $n$-vertex graphs; Yuster [13] extended this relation to packings by families of graphs. Thus fractional and integer packings have the same extremal density in the limit. A finite integer bound requires a separate argument, which we supply for triangles.

We normalize edge counts by $n^2$ and put $\rho_j = (j-1)/(2j)$, so that $t_j(n) = \rho_j n^2 - O_j(1)$. For $r \geq 3$, set $p_0 = \rho_{r-1} + 1/(4(r-1)^2)$ and, for $0 \leq p \leq \rho_r$, define

$$F_r(p) = \begin{cases} 0, & 0 \leq p \leq \rho_{r-1}, \\ p - \rho_{r-1}, & \rho_{r-1} \leq p \leq p_0, \\ \dfrac{\left(1 - \sqrt{r(r-2)(\rho_r - p)}\right)^2}{r^2}, & p_0 \leq p \leq \rho_r. \end{cases}$$

The two positive branches agree at $p_0$, where both have derivative 1. The following theorem determines the full asymptotic packing density for $K_{r+1}$-free graphs.

**Theorem 1.1.** *Let $r \geq 3$ be an integer, and let $G$ be a $K_{r+1}$-free graph on $n \geq 1$ vertices. With $p = e(G)/n^2$,*

$$\nu_r^*(G) \geq n^2 F_r(p). \tag{1}$$

*The following finite bound also holds.*

$$\nu_r^*(G) \geq e(G) - t_{r-1}(n). \tag{2}$$

*For each fixed $r$, there is a function $\varepsilon_r(n) \to 0$ such that every such graph satisfies*

$$\nu_r(G) \geq n^2 F_r(p) - \varepsilon_r(n) n^2. \tag{3}$$

*Moreover, if $m_n$ is any sequence of integers with $0 \leq m_n \leq t_r(n)$ and $m_n/n^2 \to p$, then*

$$\lim_{n\to\infty} \frac{1}{n^2} \min_{\substack{|V(G)|=n,\ e(G)=m_n \\ G \text{ is } K_{r+1}\text{-free}}} \nu_r(G) = F_r(p). \tag{4}$$

*The same limit holds with $\nu_r$ replaced by $\nu_r^*$.*

The fractional estimate is finite, whereas the integer estimate for general $r$ has an asymptotic error. For triangles this error can be removed.

**Theorem 1.2.** *Every $K_4$-free graph $G$ on $n \geq 1$ vertices satisfies*

$$\nu_3(G) \geq \left\lceil n^2 F_3\left(\frac{e(G)}{n^2}\right)\right\rceil. \tag{5}$$

*In particular, if $5n^2/16 \leq e(G) \leq n^2/3$, then*

$$\nu_3(G) \geq \left\lceil \frac{n^2}{9}\left(1-\sqrt{1-\frac{3e(G)}{n^2}}\right)^2\right\rceil.$$

*For positive integer part sizes, equality holds in* (5) *for every $K_{a,b,b}$ with $b \leq a \leq 2b$, and for every $K_{a,b,c}$ with $a = b + c$.*

*Remark* 1.3 (Comparison with known bounds). Győri and Keszegh [7] proved $\nu_3(G) \geq e(G) - \lfloor n^2/4 \rfloor$ for $K_4$-free graphs and asked for a strengthening above density $5/16$. Theorem 1.2 gives its sharp density form, since

$$F_3(p) - \left(p - \frac{1}{4}\right) = \frac{\left(1-4\sqrt{1-3p}\right)^2}{36} > 0 \qquad (5/16 < p \leq 1/3).$$

This does not determine the exact extremal value for every finite pair $(n, e)$; taking ceilings can remove a small real-valued improvement. Balogh and Wigal [2, Theorem 1.8] proved $\nu_r^*(G) \geq (2/r)(e(G) - \rho_{r-1}n^2)$ for arbitrary graphs. Theorem 1.1 determines the optimal density bound under the additional $K_{r+1}$-free assumption.

For triangles, forbidding $K_5$ in place of $K_4$ yields a piecewise linear density function. We first give a finite fractional bound with the exact Turán term.

**Proposition 1.4.** *Every $K_5$-free graph $G$ on $n$ vertices satisfies $\nu_3^*(G) \geq e(G) - \lfloor n^2/4 \rfloor$. Equality holds for $T_4(n)$ whenever 4 divides $n$.*

**Corollary 1.5.** *Let $0 \leq m_n \leq t_4(n)$ be integers with $m_n/n^2 \to p$. Then*

$$\lim_{n\to\infty} \frac{1}{n^2} \min_{\substack{|V(G)|=n,\ e(G)=m_n \\ G \text{ is } K_5\text{-free}}} \nu_3(G) = \max\{0, p - 1/4\}.$$

*The same conclusion holds for fractional packings.*

The fractional proofs start from the symmetrization lemma of Balogh and Wigal [2], which allows an arbitrary coefficient of $e(G)$ in a linear objective involving $\nu_r^*(G)$. A general symmetrization theorem for weighted fractional clique covers and decompositions appears in [1, Theorem 1.7]. These replacements preserve the forbidden-clique condition and reduce the problem to complete multipartite graphs. Under the $K_{r+1}$-free assumption, there are at most $r$ nonempty parts. With exactly $r$ parts, the fractional packing number is the product of the two smallest part sizes; with fewer parts, it is zero. Optimizing the part sizes gives the supporting lines of $F_r$. The optimizing proportions also give constructions for every prescribed sequence of feasible edge counts. For $K_5$-free

graphs, an explicit four-partite calculation gives the finite fractional triangle bound and its sharp density form.

For the finite integer triangle bound, we use the coloring lemma of Huang and Shi [11] together with the greedy-partition symmetrization of Győri and Keszegh [7]. Introducing a parameter into the triangle-counting objective gives a family of inequalities whose optimization recovers the same function $F_3$, without a rounding error. The resulting bound is attained by complete tripartite graphs. For larger cliques, a finite integer analogue can fail, as illustrated by $K_{2,2,2,2}$ in Proposition 4.1.

Section 2 proves the fractional density bounds and their asymptotic sharpness. Section 3 proves the finite integer triangle bound. Section 4 discusses finite obstructions and further questions.

## 2 Fractional packings

We use symmetrization to obtain a complete multipartite graph with no larger value of a prescribed linear packing objective. Optimizing the part sizes then gives the supporting lines of the density curve. At the end of the section, we treat $K_5$-free triangle packings by an explicit four-partite calculation.

Two nonadjacent vertices with the same neighborhood are called *clones*. We use the symmetrization lemma of Balogh and Wigal [2, Lemma 2.1] in the following form.

**Corollary 2.1.** *Let $r \geq 3$ and $s \geq 1$ be integers, let $c$ be real, and let $G$ be a $K_{s+1}$-free graph on $n$ vertices. There is a complete multipartite graph $H$ on the same vertex set, with at most $s$ nonempty parts, such that*

$$\nu_r^*(H) + ce(H) \leq \nu_r^*(G) + ce(G).$$

*Proof.* For the objective $\nu_r^* + ce$, iterate the clone-class replacements of [2, Lemma 2.1], as in the proof of [2, Lemma 2.2], to obtain a complete multipartite graph without increasing the objective. Every replacement preserves $K_{s+1}$-freeness, since a clique uses at most one vertex of the copied independent class, and replacing that vertex by a source vertex gives a clique of the same size in the previous graph. Thus the resulting graph has at most $s$ nonempty parts. □

We also use the following formula, given in [2, proof of Claim 3.2].

**Lemma 2.2.** *If $n_1 \geq \cdots \geq n_r > 0$, then*

$$\nu_r^*(K_{n_1,\ldots,n_r}) = n_{r-1}n_r.$$

*A complete multipartite graph with fewer than $r$ nonempty parts has fractional $K_r$-packing number zero.*

Merging the two smallest parts gives a finite linear bound with the exact Turán number. We record this consequence before optimizing the packing objective at higher densities.

**Proposition 2.3.** *For $r \geq 3$, every $K_{r+1}$-free graph $G$ on $n$ vertices satisfies $\nu_r^*(G) \geq e(G) - t_{r-1}(n)$.*

*Proof.* Apply Corollary 2.1 with $c = -1$, and let $H$ be the terminal graph. If $H$ has at most $r-1$ parts, then $e(H) \leq t_{r-1}(n)$ and $\nu_r^*(H) = 0$. If $H$ has $r$ parts of sizes $n_1 \geq \cdots \geq n_r$, merge its two smallest parts. The resulting $(r-1)$-partite graph has $e(H) - n_{r-1}n_r = e(H) - \nu_r^*(H)$ edges, which is at most $t_{r-1}(n)$. In both cases $\nu_r^*(H) - e(H) \geq -t_{r-1}(n)$. The reduction transfers this inequality to $G$. □

The same reduction with an arbitrary coefficient of $e(G)$ gives all the supporting lines needed for the nonlinear density bound.

**Proposition 2.4.** *For $r \geq 3$ and $\lambda \geq 1$, put $D = r-2+r\lambda$ and $B_r(\lambda) = \lambda((r-3)+(r-1)\lambda)/(2D)$. Every $K_{r+1}$-free graph $G$ on $n$ vertices satisfies*

$$\nu_r^*(G) \geq \lambda e(G) - B_r(\lambda)n^2. \tag{6}$$

*For $\lambda > 1$, equality in the continuous optimization is attained at the part proportions*

$$\underbrace{\frac{1+\lambda}{D}, \ldots, \frac{1+\lambda}{D}}_{r-2 \text{ parts}}, \quad \frac{\lambda}{D}, \quad \frac{\lambda}{D}. \tag{7}$$

*Proof.* Apply Corollary 2.1 with $c = -\lambda$. It is enough to prove (6) for complete multipartite graphs with at most $r$ nonempty parts. If there are at most $r-1$ parts, then $e(G) \leq \rho_{r-1}n^2$, and

$$B_r(\lambda) - \lambda\rho_{r-1} = \frac{\lambda(\lambda-1)}{2(r-1)D} \geq 0.$$

This proves the required bound in that case.

Suppose there are $r$ parts. Normalize their sizes by $n$, write $y, z$ for the two smallest proportions, and put $s = y+z$. Denote the other $r-2$ proportions by $x_1, \ldots, x_{r-2}$. Lemma 2.2 gives $\nu_r^*(G)/n^2 = yz$. Since $\sum x_i = 1-s$, the Cauchy–Schwarz inequality gives $\sum_{i<j} x_i x_j \leq (r-3)(1-s)^2/(2(r-2))$. Also, $yz \leq s^2/4$ and $1-\lambda \leq 0$. Hence

$$\begin{aligned}\frac{\nu_r^*(G) - \lambda e(G)}{n^2} &= (1-\lambda)yz - \lambda s(1-s) - \lambda \sum_{i<j} x_i x_j \\ &\geq \frac{1-\lambda}{4}s^2 - \lambda s(1-s) - \frac{\lambda(r-3)}{2(r-2)}(1-s)^2 = \frac{D}{4(r-2)}\left(s - \frac{2\lambda}{D}\right)^2 - B_r(\lambda).\end{aligned}$$

The nonnegative square proves (6). When $\lambda > 1$, equality in the two inequalities and in the square is attained by (7). These proportions are ordered as required, since $1+\lambda > \lambda$. □

Optimizing the supporting lines gives the density function, and fractional rounding then yields the asymptotic integer bound.

*Proof of the lower bounds in Theorem 1.1.* The bound $\nu_r^*(G) \geq 0$ handles $p \leq \rho_{r-1}$. At $\lambda = 1$, Proposition 2.4 gives $B_r(1) = \rho_{r-1}$ and thus $\nu_r^*(G) \geq n^2(p - \rho_{r-1})$. For $p_0 \leq p < \rho_r$, put

$$z = \sqrt{r(r-2)(\rho_r - p)}, \qquad \lambda = \frac{(r-2)(1-z)}{rz}. \tag{8}$$

The condition $p \geq p_0$ is equivalent to $z \leq (r-2)/(2(r-1))$, so $\lambda \geq 1$. Substituting (8) into (6) gives

$$\lambda p - B_r(\lambda) = \frac{(1-z)^2}{r^2} = F_r(p).$$

At $p = \rho_r$, taking $\lambda \to \infty$ in (6) gives $\nu_r^*(G) \geq n^2/r^2$. This proves (1) on the entire feasible interval. Proposition 2.3 proves (2).

The theorem of Haxell and Rödl [8], applied to $K_r$, states that for each fixed $r$ and each $\varepsilon > 0$, every sufficiently large graph $G$ on $n$ vertices satisfies $\nu_r^*(G) - \nu_r(G) \leq \varepsilon n^2$. The threshold is uniform over $G$. Applying this estimate to (1) proves (3). □

To prove sharpness, we describe complete multipartite graphs attaining every branch of the density function. Rounding their part sizes then gives the statement for every feasible sequence of edge counts, rather than only for rational densities along subsequences.

For $p \le \rho_{r-1}$, complete $(r-1)$-partite graphs with proportions $1-(r-2)x, x, \dots, x$, where $0 \le x \le 1/(r-1)$, realize every density between 0 and $\rho_{r-1}$ and contain no $K_r$. For $\rho_{r-1} \le p \le p_0$, take $r-2$ parts of proportion $1/(r-1)$ and two further parts of proportions $y, z$ with $y+z = 1/(r-1)$ and $yz = p - \rho_{r-1}$. The conditions on $p$ ensure that such nonnegative $y, z$ exist. They are the two smallest proportions. The edge density is $\rho_{r-1} + yz = p$, and Lemma 2.2 gives fractional packing density $yz = F_r(p)$; a zero part corresponds to a $K_r$-free graph.

For $p_0 \le p < \rho_r$, use (7) with the parameter in (8). The last two proportions are $(1-z)/r$. The other $r-2$ proportions are $(1 + 2z/(r-2))/r$, so their squared sum gives edge density $\rho_r - z^2/(r(r-2)) = p$. The fractional packing density is $(1-z)^2/r^2 = F_r(p)$. At $p = \rho_r$, all $r$ proportions are $1/r$.

*Proof of the sharpness assertion in Theorem 1.1.* Fix a feasible sequence $m_n$ with $m_n/n^2 \to p$. The lower bounds already proved and the continuity of $F_r$ give the required lower limit for both packing parameters. It remains to construct graphs giving the upper limit.

If $p < \rho_{r-1}$, then $m_n \le t_{r-1}(n)$ for all sufficiently large $n$. Choose any $m_n$ edges of $T_{r-1}(n)$. The resulting graph has no $K_r$ and therefore has both packing parameters equal to zero.

Suppose $\rho_{r-1} \le p < \rho_r$, and set $\widehat{p}_n = \max\{m_n/n^2, \rho_{r-1}\} + n^{-1/2}$. For all sufficiently large $n$, this lies in $[\rho_{r-1}, \rho_r)$ and tends to $p$. Choose the part proportions just described for density $\widehat{p}_n$, and round their multiples by $n$ to nonnegative integers with sum $n$. Each part changes by at most a constant depending only on $r$. The resulting complete multipartite graph $H_n$ has

$$e(H_n) = \widehat{p}_n n^2 + O_r(n), \qquad \nu_r^*(H_n) = F_r(\widehat{p}_n)n^2 + O_r(n).$$

The first quantity is at least $m_n$ for all sufficiently large $n$, because the added density contributes $n^{3/2}$ edges. Delete $e(H_n) - m_n = o(n^2)$ edges. This does not introduce a $K_{r+1}$ and cannot increase either packing parameter. The resulting graph $G_n$ satisfies

$$\nu_r(G_n) \le \nu_r^*(G_n) \le F_r(\widehat{p}_n)n^2 + O_r(n) = \big(F_r(p) + o(1)\big)n^2.$$

This proves the upper limit in the interior, including $p = \rho_{r-1}$ and $p = p_0$.

Finally, if $p = \rho_r$, delete $t_r(n) - m_n = o(n^2)$ edges from $T_r(n)$. Lemma 2.2 gives $\nu_r^*(T_r(n)) = n^2/r^2 + O_r(n)$. Monotonicity under edge deletion therefore gives the upper limit $F_r(\rho_r) = 1/r^2$. This completes the proof of (4) and its fractional counterpart. □

We now allow four parts for triangle packings. The following formula provides both the finite bound in Proposition 1.4 and the constructions for Corollary 1.5.

**Lemma 2.5.** *For positive integers $a \le b \le c \le d$, $\nu_3^*(K_{a,b,c,d}) = \min\{ad + bc, ab + ac + bc\}$.*

*Proof.* Denote the parts by $A, B, C, D$ in the indicated order. The edges between $A, D$ together with those between $B, C$ cover every triangle. So do the edges between the three parts $A, B, C$. These covers give the two upper bounds.

For the lower bound, prescribe the total weights of triangle types $ABC, ABD, ACD, BCD$, distributing each total uniformly within its type. If $d \ge b + c$, use the totals $(0, ab, ac, bc)$. The loads on $AB, AC, BC$ equal their capacities, while those on $AD, BD, CD$ are at most $ad, bd, cd$, respectively, since $a(b+c) \le ad$, $b(a+c) \le bd$ and $c(a+b) \le cd$. The total weight is $ab + ac + bc$.

If $d \le b + c$, put $x = a(b + c - d)/2$ and $y = a(b + d - c)/2$ and use $(x, y, ad - y, bc - x)$. These totals are nonnegative by the ordering of the part sizes. The loads on $AB, AC, AD, BC$ equal their

capacities. The other two loads are $a(d-c)+bc \le bd$ and $a(d-b)+bc \le cd$. This gives total weight $ad+bc$, as required. □

*Proof of Proposition 1.4.* For a complete four-partite graph as in Lemma 2.5,

$$e(G) - \nu_3^*(G) = \max\{(a+d)(b+c), d(n-d)\} \le \lfloor n^2/4 \rfloor.$$

Graphs with at most three parts satisfy the same bound by Proposition 2.3. Corollary 2.1 with $s=4$ and $c=-1$ transfers it to every $K_5$-free graph. When all four parts have size $t$, Lemma 2.5 gives $\nu_3^* = 2t^2 = e - (4t)^2/4$, proving the equality assertion. □

*Proof of Corollary 1.5.* The lower bounds follow from Proposition 1.4, nonnegativity, and the Haxell–Rödl theorem. For $1/4 \le p \le 3/8$, take four part proportions $x, x, 1/2-x, 1/2-x$, where $0 \le x \le 1/4$. Their edge density is $p = 1/4 + x - 2x^2$, and Lemma 2.5 gives packing density $x - 2x^2 = p - 1/4$ when $x > 0$. At $x = 0$, the graph is complete bipartite and has no triangle. These proportions cover the indicated interval continuously.

For a sequence with $1/4 \le p < 3/8$, choose these proportions at density $\widehat{p}_n = \max\{m_n/n^2, 1/4\} + n^{-1/2}$ and round the part sizes. For all sufficiently large $n$ the resulting $K_5$-free graph has $\widehat{p}_n n^2 + O(n) \ge m_n$ edges and fractional packing number $(\widehat{p}_n - 1/4)n^2 + O(n)$. Deleting edges to leave exactly $m_n$ cannot increase either packing number. If $p < 1/4$, take any $m_n$ edges of $T_2(n)$ for all sufficiently large $n$. At $p = 3/8$, delete $t_4(n) - m_n = o(n^2)$ edges from $T_4(n)$. These constructions prove the required upper limits in all cases. □

## 3 Integer packings

The fractional density theorem gives an integer bound with an additive $o(n^2)$ error. For $K_4$-free graphs, a triangle-counting argument removes this error. A coloring lemma converts a suitable lower bound on the total number of triangles directly into a packing. We obtain that lower bound by inserting a parameter into the greedy-partition symmetrization of Győri and Keszegh [7].

**Definition 3.1.** A *greedy clique partition* of a graph $G$ with at least one vertex is a partition $P = (C_0, \dots, C_h)$ of $V(G)$ into nonempty cliques such that $|C_0| \le \cdots \le |C_h|$ and $G[C_0 \cup \cdots \cup C_i]$ is $K_{|C_i|+1}$-free for every $i$. Its size is $q(P) = h+1$.

Every graph with at least one vertex has such a partition. Repeatedly remove a maximum clique from the remaining graph and then order the removed cliques in reverse order. The maximum-clique condition gives both the size order and the forbidden-clique condition in the definition.

Write $t(G)$ for the total number of triangles in $G$. We use the coloring lemma of Huang and Shi [11], in the form stated in [7, arXiv version, Lemma 4].

**Lemma 3.2** (Huang–Shi)**.** *If $G$ is $K_4$-free and $P$ is a greedy clique partition of $G$, then*

$$\nu_3(G) \ge \frac{t(G)}{q(P)}.$$

The next lemma supplies all the supporting lines needed for the finite triangle bound. At $\lambda = 1$ it is the triangle-count estimate of Győri and Keszegh [7, arXiv version, Theorem 5]. The proof follows their algorithm, with the parameter and its positive-region condition made explicit. Their concluding remarks also describe the use of related potential functions.

**Lemma 3.3.** *Let $G$ be a $K_4$-free graph on $n \geq 1$ vertices, let $P$ be a greedy clique partition, and let $\lambda \geq 1$. Then*

$$t(G) \geq q(P)\left(\lambda e(G) - \frac{\lambda^2}{1+3\lambda}n^2\right).$$

*Proof.* Fix $\lambda \geq 1$, put $B = \lambda^2/(1+3\lambda)$, and suppose that $\Psi(G,P) = q(P)(\lambda e(G) - Bn^2) - t(G) > 0$. We use the generalized greedy partitions and the matching and merging procedure of [7, arXiv version, Section 2]. For such a partition, $q(P)$ is the size of the largest independent part of the first block plus the number of remaining clique blocks. The structural properties proved there ensure that each round absorbs the second block into the first, does not decrease $q(P)$, and preserves the partition requirements and the clique-number bound. We verify the change of objective needed here.

During one round, fix $q_0 = q(P)$. The objective $q_0(\lambda e - Bn^2) - t$ is linear in $e$ and $t$. For a vertex $v$, let $d(v)$ and $t_v$ denote its degree and its triangle count in the current graph, and use the local score $q_0\lambda d(v) - t_v$ in the clone replacements. The contribution of two nonadjacent clone classes is the sum of their sizes times their respective scores, plus a term unchanged by the replacement. Copying the class with larger score therefore does not decrease the objective. The fixed-parameter symmetrization in [7, arXiv version, Sections 2–3] therefore gives $q_0(\lambda e(G') - Bn^2) - t(G') > 0$ at the end of the round. Since $t(G') \geq 0$ and $q_0 > 0$, this implies $\lambda e(G') - Bn^2 > 0$. Thus replacing $q_0$ by $q(P') \geq q_0$ also does not decrease the objective, and the potential remains positive.

The procedure terminates at a complete multipartite graph. By $K_4$-freeness, write its part sizes as $a \geq b \geq c \geq 0$, with $a > 0$. Here $q = a$, $t = abc$, $e = a(n-a) + bc$, and $b + c = n - a$. Since $\lambda \geq 1$,

$$\begin{aligned}\frac{t}{q} - \lambda e &= (1-\lambda)bc - \lambda a(n-a) \geq \frac{1-\lambda}{4}(n-a)^2 - \lambda a(n-a) \\ &= \frac{1+3\lambda}{4}\left(a - \frac{1+\lambda}{1+3\lambda}n\right)^2 - Bn^2 \geq -Bn^2.\end{aligned}$$

The terminal potential is therefore nonpositive, a contradiction. □

*Proof of the lower bound in Theorem 1.2.* Choose a greedy clique partition $P$ of $G$. Lemmas 3.2 and 3.3 give, for every $\lambda \geq 1$,

$$\nu_3(G) \geq \frac{t(G)}{q(P)} \geq \lambda e(G) - \frac{\lambda^2}{1+3\lambda}n^2.$$

The supporting-line coefficient here is exactly $B_3(\lambda)$. Taking $\lambda = 1$ gives $e(G) - n^2/4$, and choosing (8) with $r = 3$ gives the upper branch of $n^2 F_3(e(G)/n^2)$. The endpoint $p = 1/3$ follows by taking $\lambda \to \infty$. Together with $\nu_3(G) \geq 0$, these bounds cover all feasible densities. Since $\nu_3(G)$ is an integer, taking the ceiling proves (5). □

For the equality cases, we use the standard identity for positive integers $a \geq b \geq c$

$$\nu_3(K_{a,b,c}) = bc. \tag{9}$$

It follows from Lemma 3.2 and the fact that a greedy partition of $K_{a,b,c}$ has $a$ cliques [7, arXiv version, Observation 7], together with the upper bound from the edges between the two smallest parts.

*Proof of the equality assertions in Theorem 1.2.* Consider $K_{a,b,b}$ with $b < a \leq 2b$ and set $\lambda = \frac{b}{a-b} \geq 1$. Its normalized part sizes agree with (7) for $r = 3$, so equality holds in the supporting line and

in the optimized density bound. Identity (9) gives $\nu_3 = b^2 = n^2 F_3(e/n^2)$. If $a = b$, the graph is balanced and the same equality follows at $p = 1/3$.

If $a = b + c$, reorder $b, c$ so that $b \geq c$. Then $a = n/2$, $e = n^2/4 + bc$, and $bc \leq n^2/16$. This density belongs to the linear branch, and (9) gives $\nu_3 = bc = n^2 F_3(e/n^2)$. In each case the expression before taking the ceiling is already an integer. □

# 4 Concluding remarks

Theorem 1.1 determines the full asymptotic $K_r$-packing density in $K_{r+1}$-free graphs, and Corollary 1.5 gives the full triangle-packing density in $K_5$-free graphs. For triangles with a forbidden $K_{M+1}$, the complete density function remains open for $M \geq 5$. For $r \geq 4$, the corresponding problem already remains open for $K_{r+2}$-free graphs. Sharp linear bounds do not by themselves determine these functions at intermediate densities. The optimal uniform linear coefficient for all $K_r$-packings and forbidden-clique orders, together with its fixed-forbidden-subgraph consequences, is determined in a separate paper [9]. The linear coefficient results in an earlier publicly circulated version of the present paper have been incorporated into that general treatment. The proofs here are independent of the general coefficient theorem.

For integer triangle packings, $K_4$ already shows that $\nu_3(G) \geq e(G) - t_2(n)$ fails for $K_5$-free graphs. More generally, let $G = T_4(4a)$ with $a$ odd. Every vertex has odd degree, whereas the edges covered by triangles have even degree at each vertex. At least $n/2$ edges must remain uncovered, giving $\nu_3(G) \leq e(G)/3 - n/6 = e(G) - n^2/4 - n/6$. Thus a uniform finite extension of the asymptotic bound would require an error of at least linear order. Whether an $O(n)$ error always suffices is a natural further question.

The finite triangle bound raises the question of whether the asymptotic error in Theorem 1.1 can also be removed for larger cliques. The following example shows that the analogous finite integer statement fails already for $r = 4$.

**Proposition 4.1.** *The graph $G = K_{2,2,2,2}$ is $K_5$-free and satisfies*

$$|V(G)| = 8, \qquad e(G) = 24, \qquad \nu_4(G) = 2, \qquad \nu_4^*(G) = 4.$$

*In particular, neither $\nu_4(G) \geq n^2 F_4(e(G)/n^2)$ nor $\nu_4(G) \geq e(G) - t_3(n)$ holds for every $K_5$-free graph.*

*Proof.* Label the two vertices in each part by 0 and 1. Each $K_4$ corresponds to a word in $\{0,1\}^4$. Two such cliques are edge-disjoint if and only if the words agree in at most one coordinate, or equivalently have Hamming distance at least 3.

Suppose three words have pairwise distance at least 3. Flipping coordinates makes the first word 0000 without changing the distances. Each of the other two words then has at least three entries equal to 1, so their distance is at most 2, a contradiction. The words 0000 and 1111 give a packing of size 2.

Lemma 2.2 gives $\nu_4^*(G) = 4$. Finally, $e(G)/n^2 = 3/8 = \rho_4$, hence $n^2 F_4(e(G)/n^2) = 4$, whereas $e(G) - t_3(8) = 24 - 21 = 3$. Both proposed integer bounds exceed $\nu_4(G) = 2$. □

Theorem 1.1 determines the leading term at each fixed feasible density, and Theorem 1.2 gives a finite bound attained by explicit families. Further questions include a characterization of equality in (5), a stability theorem above the transition density, and the smallest possible integer error for $r \geq 4$.

A separate question concerns a specified greedy partition $P$ of size $q$. He, Ma, Wang, and Zu [10] disproved the proposed triangle-count inequality $t(G) \geq q(e(G) - q(n-q))$ from [7]. As they note in their concluding remarks, the corresponding packing assertion $\nu_3(G) \geq e(G) - q(n-q)$ remains open. Our bound depends only on $n$ and $e(G)$ and does not establish that partition-dependent assertion.

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